ERRATA CORRIGE

Francesco Zonta, Ph.D.

Associate Professor

Department of Biosciences and Bioinformatics

Room SB465, francesco.zonta@xjtlu.edu.cn



Xi'an Jiaotong-Liverpool University

西交利物浦大學

The force is the gradient of the potential energy

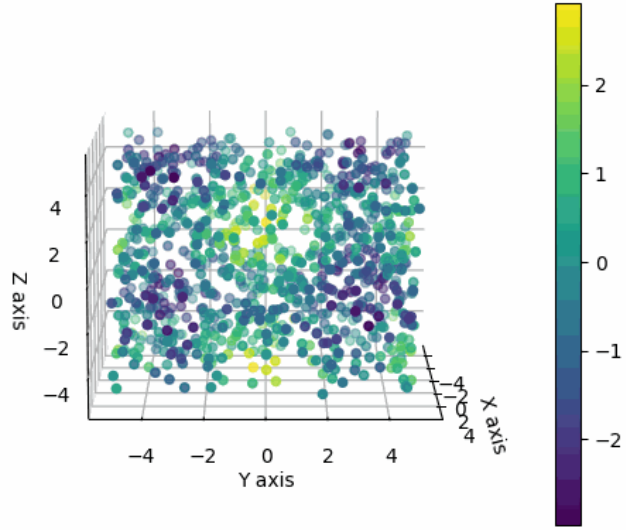
In general, the potential energy does not depend on a single coordinate but is a function of the position in space:

$$U = U(x, y, z)$$

In this case, the force is given by the opposite of the *gradient* of the potential energy:

$$\boldsymbol{F} = -\boldsymbol{\nabla U} = \left(-\frac{\partial U}{\partial x}, -\frac{\partial U}{\partial y}, -\frac{\partial U}{\partial z}\right)$$

In other words, an object experiencing an energy potential, feels a force in the direction in which the energy potential is decreasing most rapidly.

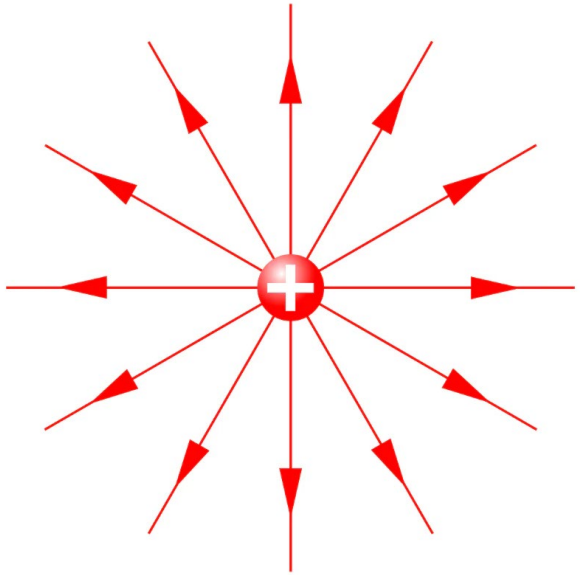


Example of a scalar field in a 3-dimensional space.

The temperature in a room, can have slightly different values depending on the position in space.

Analogously, the potential energy of an object has different values depending on its position in space.

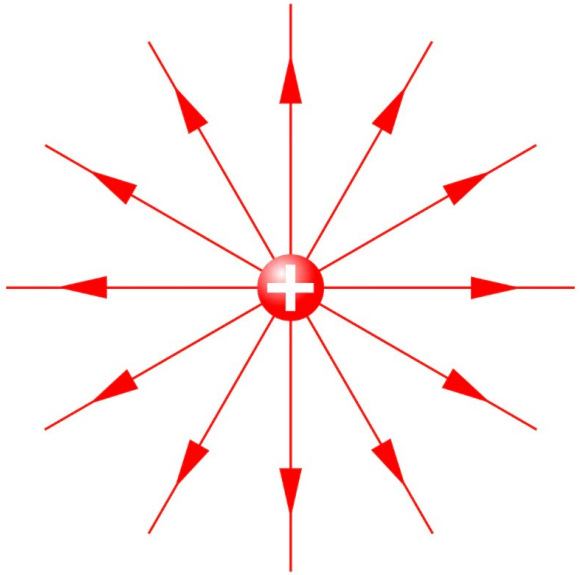
The electrostatic potential

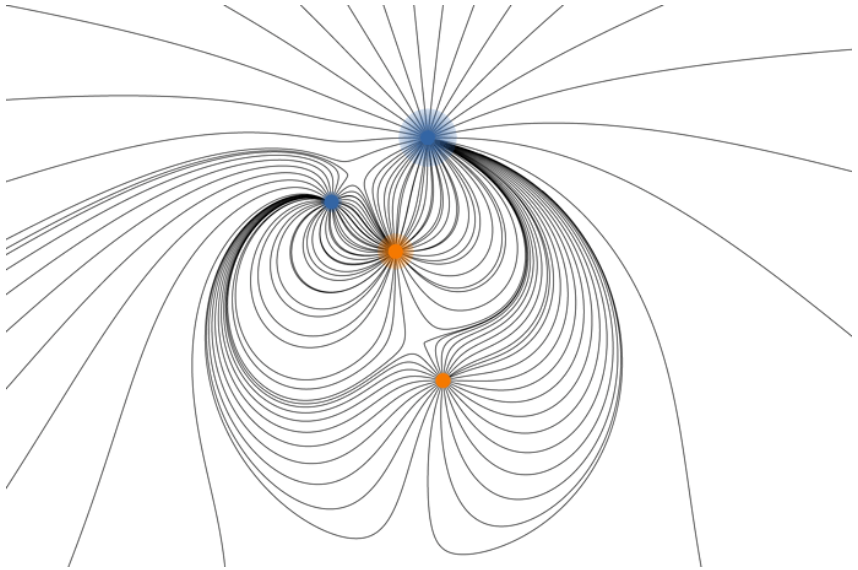


The potential energy of a charge q, which is interacting with another charge Q is proportional to q.

$$\boldsymbol{F} = k\frac{qQ}{r_{12}^2}\boldsymbol{u}_{r_{12}} \Rightarrow U = +k\frac{qQ}{r_{12}} = q\left(+k\frac{Q}{r_{12}}\right)$$

The electrostatic potential





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This is true in general, the potential energy felt by a charge q is proportional to the charge itself (even if in general the field can be generated by many charges and have a very complicated form).

Similarly to the definition of the electric field, we can now define the electrostatic potential as the ration between the electrostatic potential energy and the charge that feels that energy:

$$V(\boldsymbol{r}) = \frac{U(\boldsymbol{r})}{q} \qquad \boldsymbol{E} = -\boldsymbol{\nabla V}$$

2025-26 S1 – BIO220 PROGRAM

Week	Lecture	Tutorials and Labs	Coursework
Week 1	Introduction to computational Biology.		
Week 2	Charges and Coulomb law. Electrostatic potential.		
Week 3	Atoms and molecules. The Schrodinger equation		
Week 4	Molecular mechanics. Force Fields.		Coursework 1 (in class test): Molecular Physics.
Week 5	Structure of proteins and nucleic acids.	Tutorial and Lab: Molecular visualization with VMD.	
Week 6	Structural models and protein folding. Homology modeling and Alphafold.	Tutorial and Lab: Homology modeling.	Coursework 2 (laboratory): Homology modeling and molecular visualization.
Week 7	Molecular surfaces and electrostatics.		
Week 8	Molecular docking.		Coursework 2 deadline
Week 9	Energy minimization. Normal modes analysis.	Tutorial and Lab: Molecular docking, normal modes.	
Week 10	Free energy and solvation.		
Week 11	Small molecules and chemical spaces. Chemoinformatics.	Tutorial and lab: Protein/Ligand modeling.	Coursework 3 (laboratory): Protein/Ligand modeling.
Week 12	Drug libraries, screening and selection.		
Week 13	Recap.		Coursework 3 deadline

THE SCHROEDINGER EQUATION AND MOLECULAR PHYSICS

Francesco Zonta, Ph.D.

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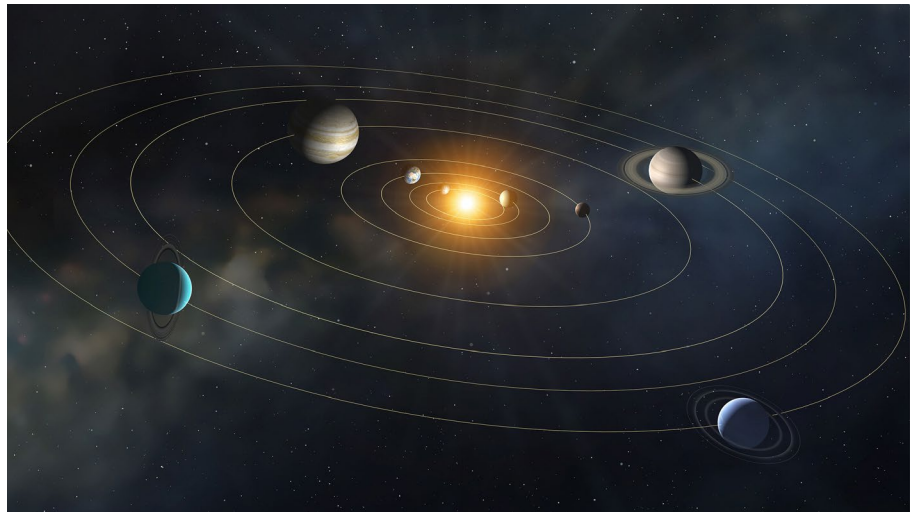
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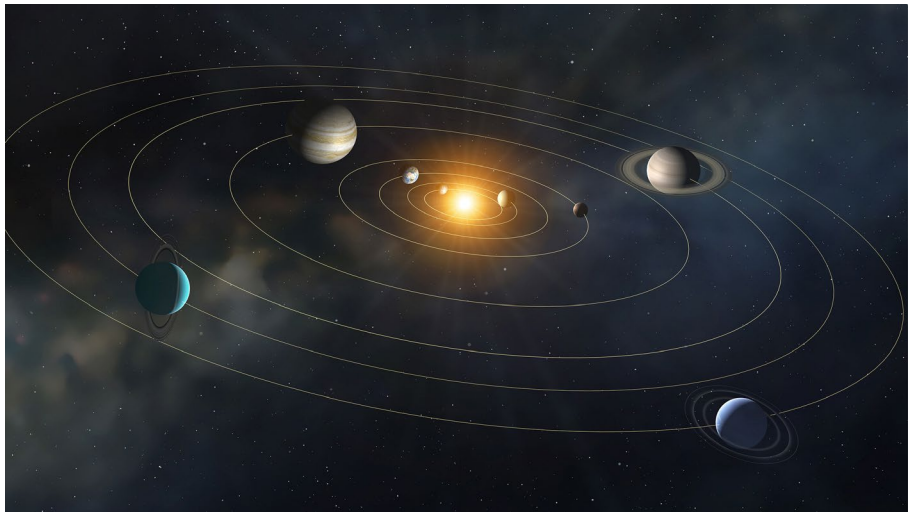
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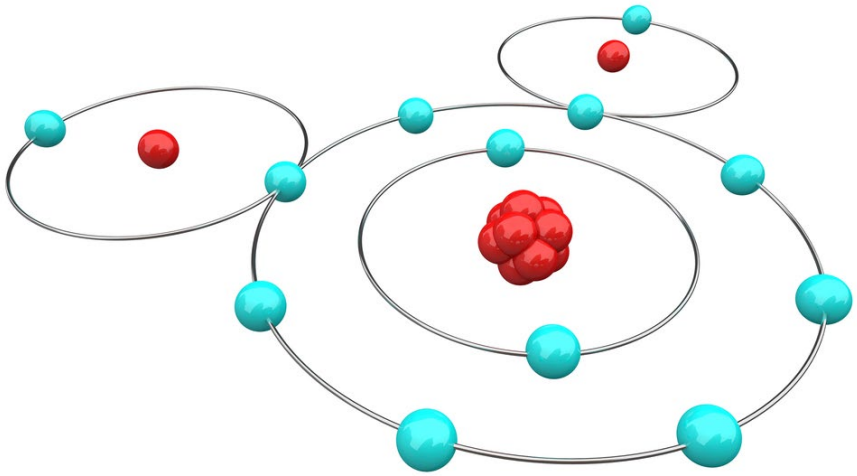
Biology from the point of view of a physicist



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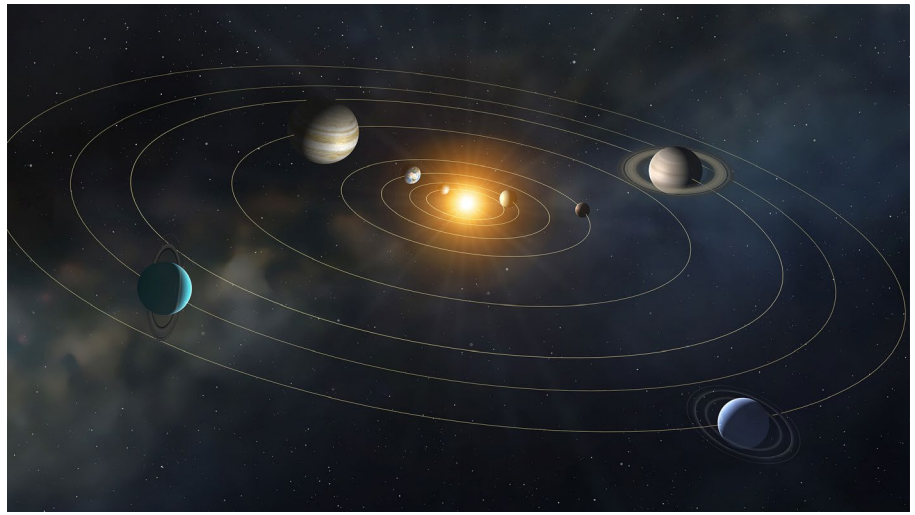
Biology from the point of view of a physicist

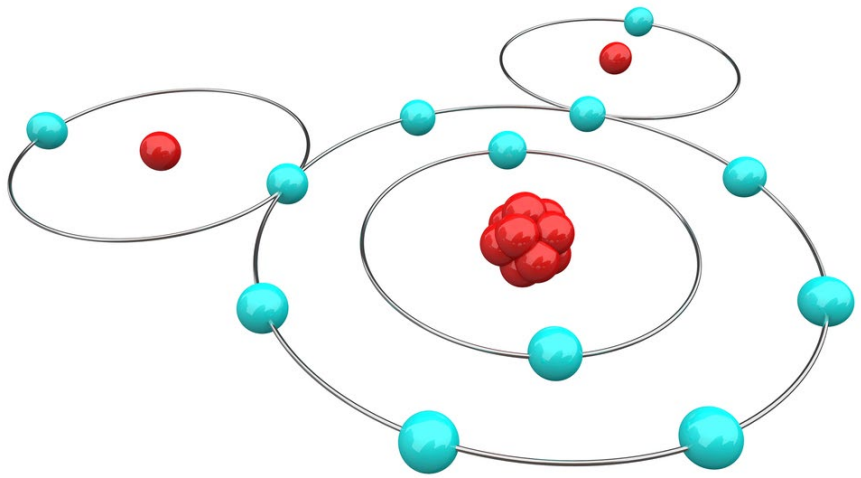




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Biology from the point of view of a physicist





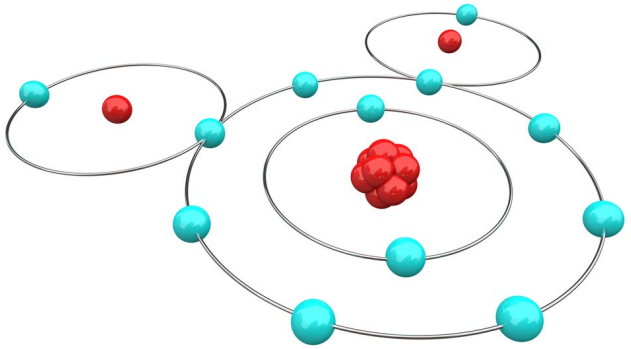
We want to describe our biological system of interest from the point of view of the Physics and derive its properties from physical principles.

From a theoretical point of view, a biological system can be regarded as a collection of nuclei and electrons, interacting through the electrostatic potential.

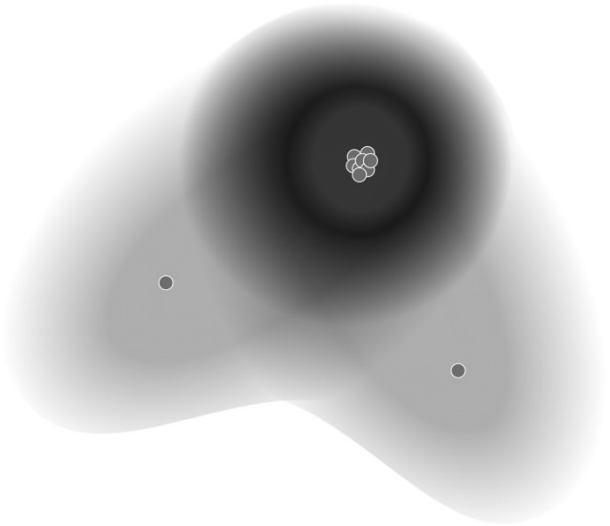
$$F = -G\frac{Mm}{r^2} \qquad F = k\frac{q_1 q_2}{r^2}$$

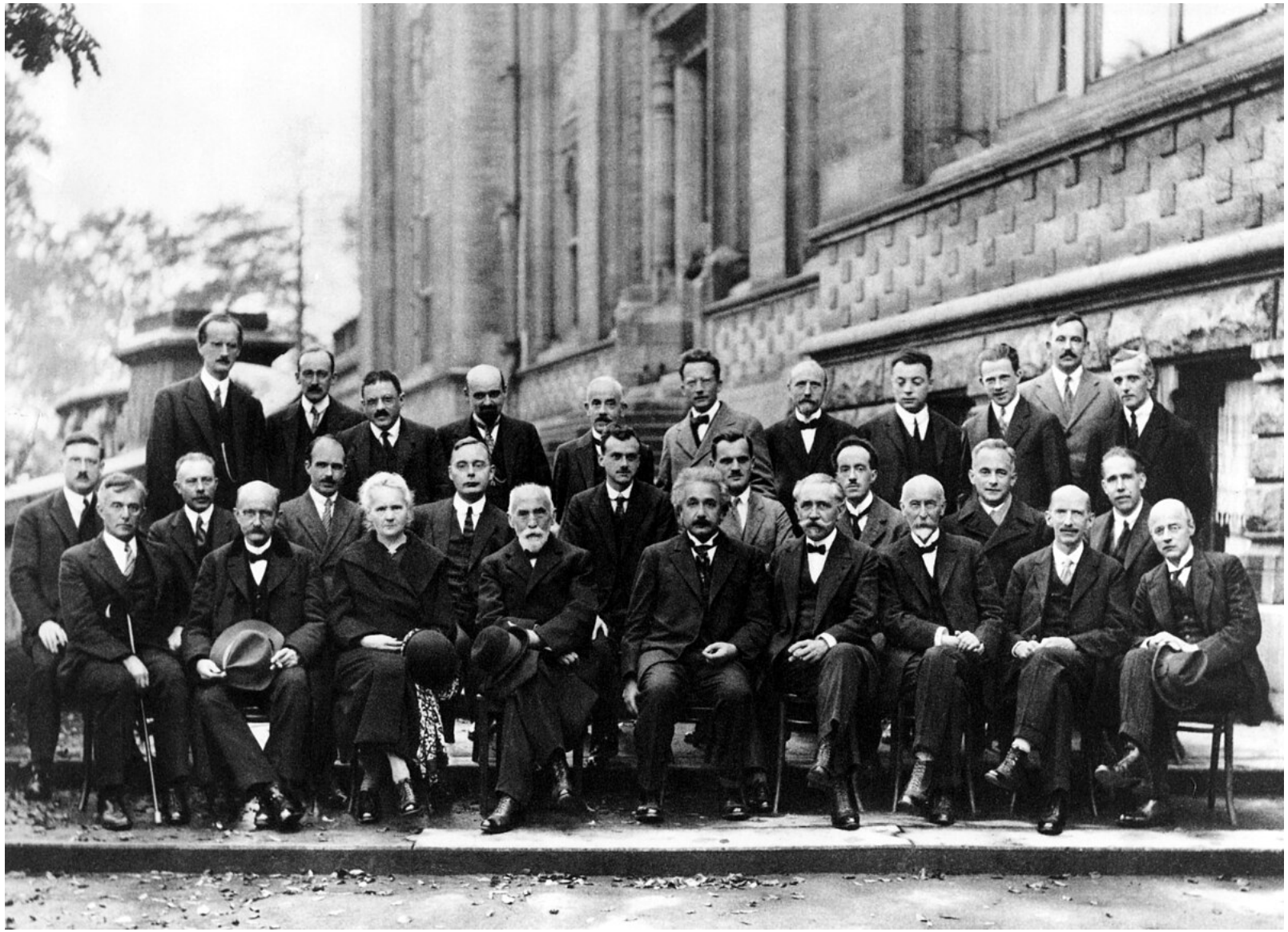
From the mathematical point of view, the electrostatic potential and the gravitational potential are very similar. But atoms are very different from solar system and galaxies are very different from cells…

Quantum Mechanics, Atoms and Molecules

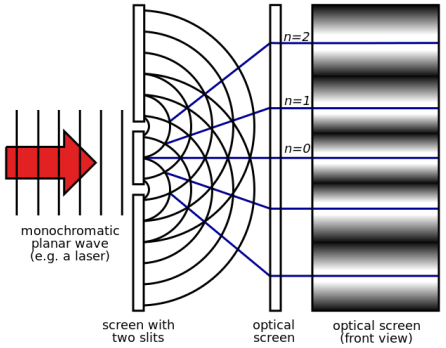


QM

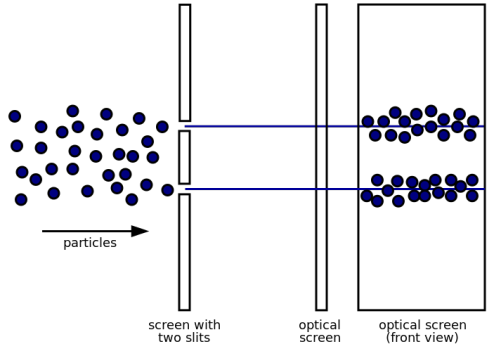




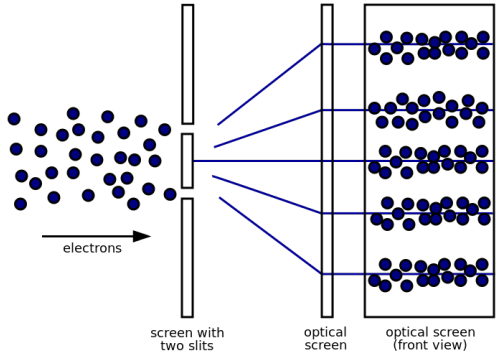
The double slit experiment



Classical waves produce a diffraction pattern due to constructive and destructive interference.



Classical particles pass through one of the slit. They produce a signal with two peaks on the other side of the screen.



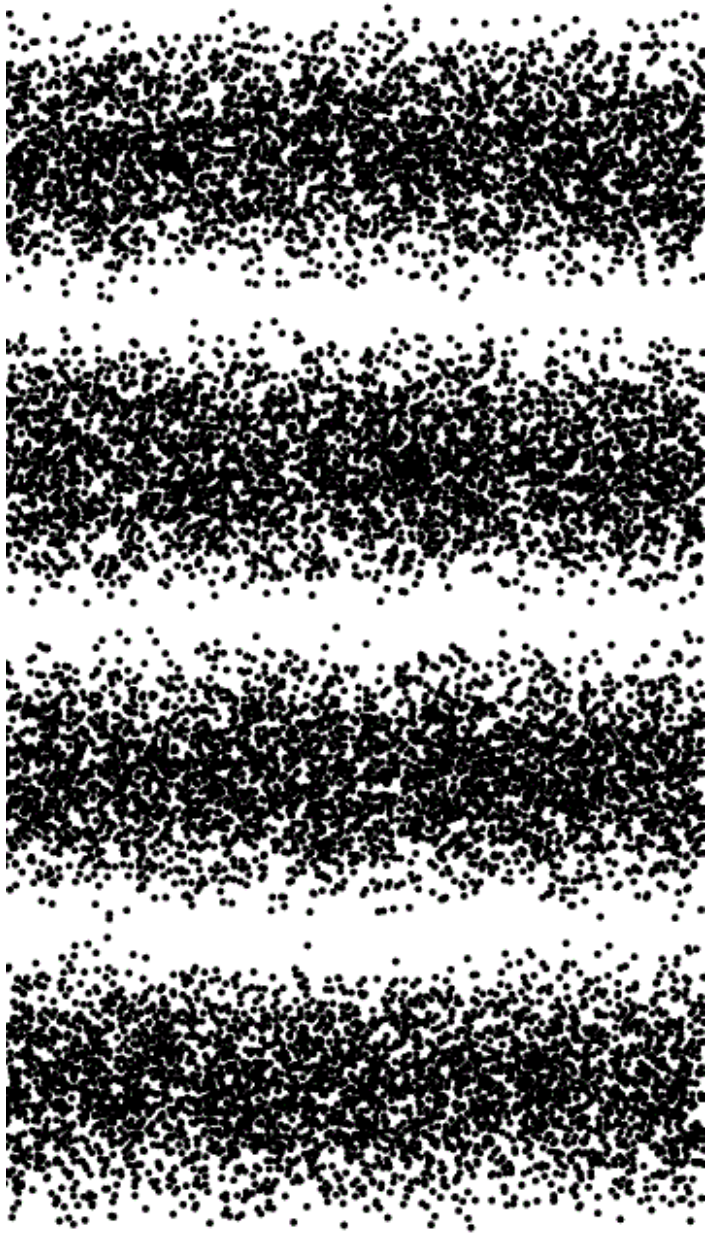
Particle–wave duality: quantum particles can behave as classical particles or as waves, depending on how we measure their characteristics.

Particle – wave duality

Each electron taken alone can behave as a particle [1], in the sense that it will be revealed by a detector at a precise location.

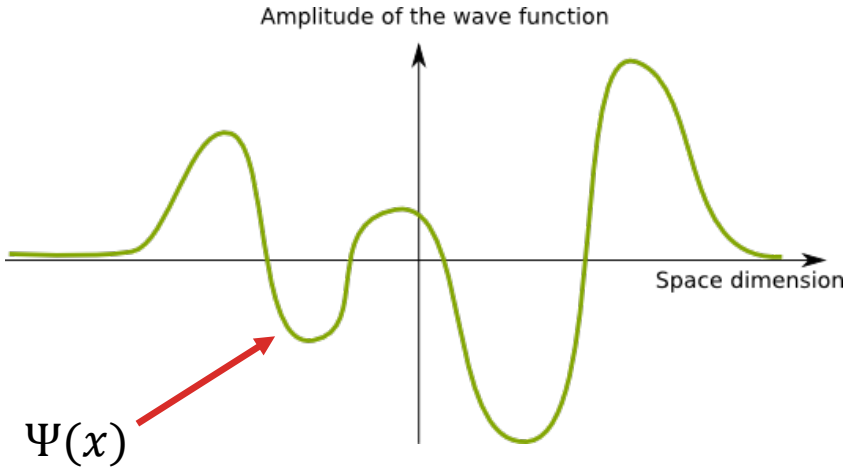
But the pattern of where all the electrons land shows wave behavior - instead of just two bands behind the slits, we see multiple bands with dark and bright regions.

Note: The same effect applies to all objects - photons, atoms, even molecules. But for particles as massive as a proton (or more massive than it), the wave behavior is negligible.



[1] *not really… but let's say so, for the moment*

Wave functions

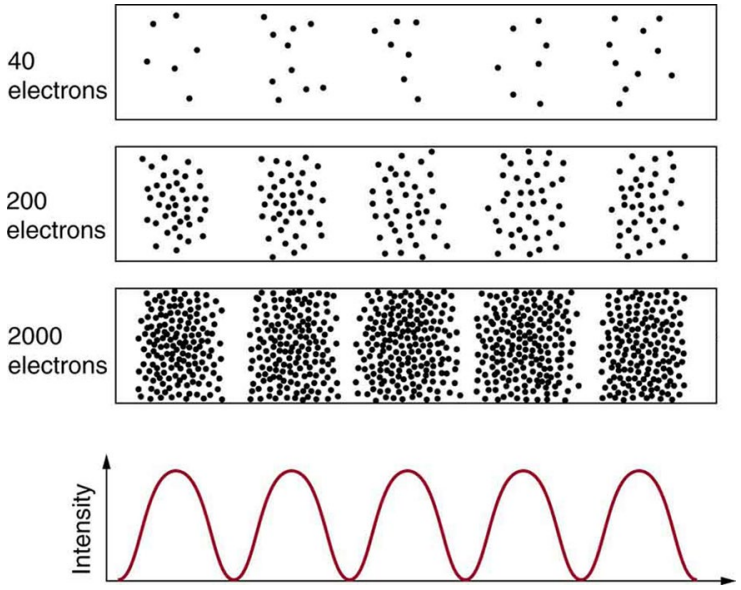


To describe this behavior, we introduce the mathematical concept of wave function.

$\Psi(x)$: wave function.
It represent the quantum state of the system

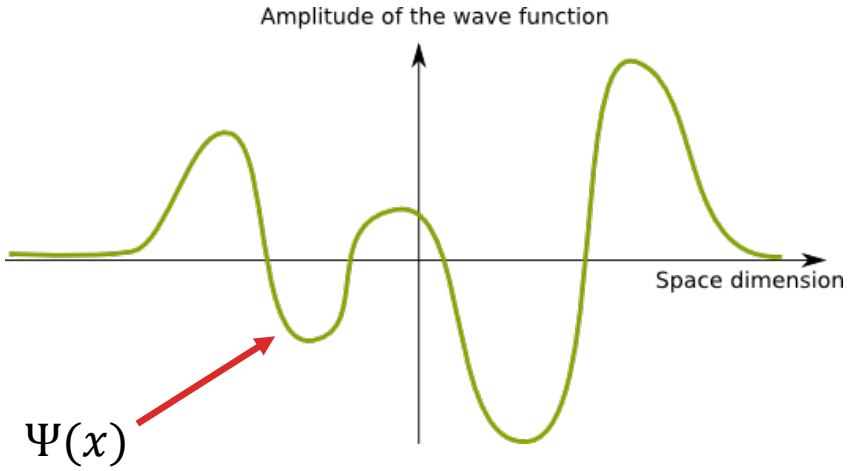
The function $\Psi(x)$ can be used to determine the probability that a particle (i.e. an electron, e) is found in a certain part of the space (i.e the interval [a,b]).

$$P(e \in [a,b]) = \int_a^b |\Psi(x)| dx$$



Mathematically, the wave function is a complex function.
This is way we take the modulus squared of $\Psi(x)$.

Wave functions



The wave function is normalized to 1
(the electron must be somewhere!)

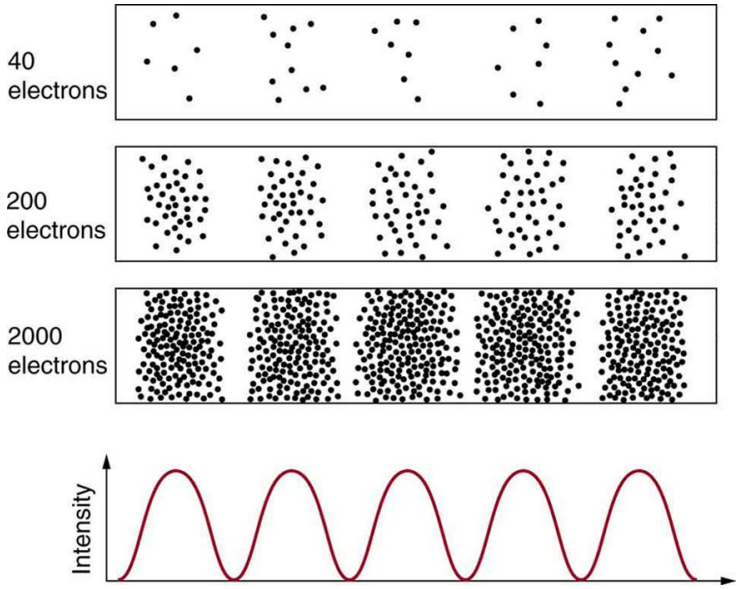
$$\int_{-\infty}^{+\infty} |\Psi(x)|^2 dx = 1$$

If $\Psi_1(x)$ and $\Psi_2(x)$ are possible states of a system, their linear combinations (normalized to 1) are still possible states of the system:

$$\Psi(x) = c_1\Psi_1(x) + c_2\Psi_2(x)$$

$$c_1, c_2 \epsilon \mathbb{C}$$

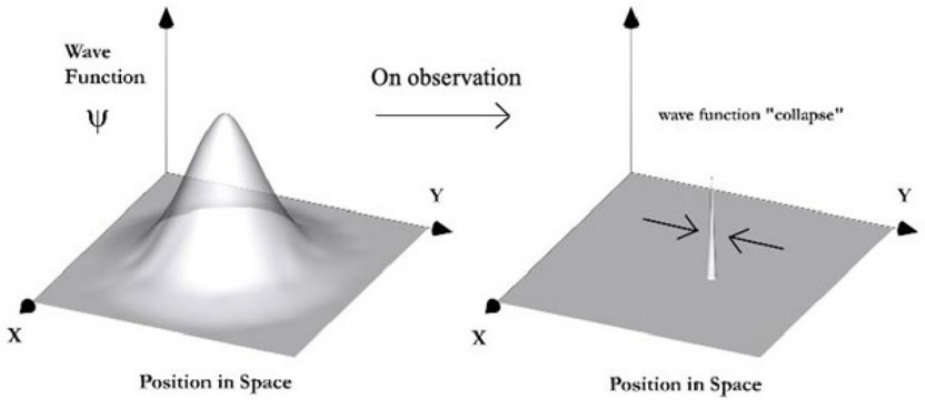
$$|c_1|^2 + |c_2|^2 = 1$$



Measurements and collapse of the wave function

When we take a measurement (for example the position of an electron), we do not get the wave function, but we get the result of the measurement (for example the coordinates x,y,z).

Measurements and collapse of the wave function

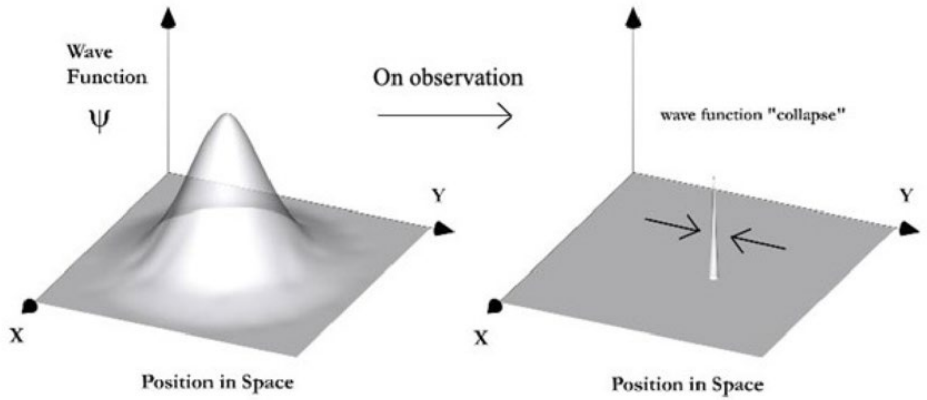


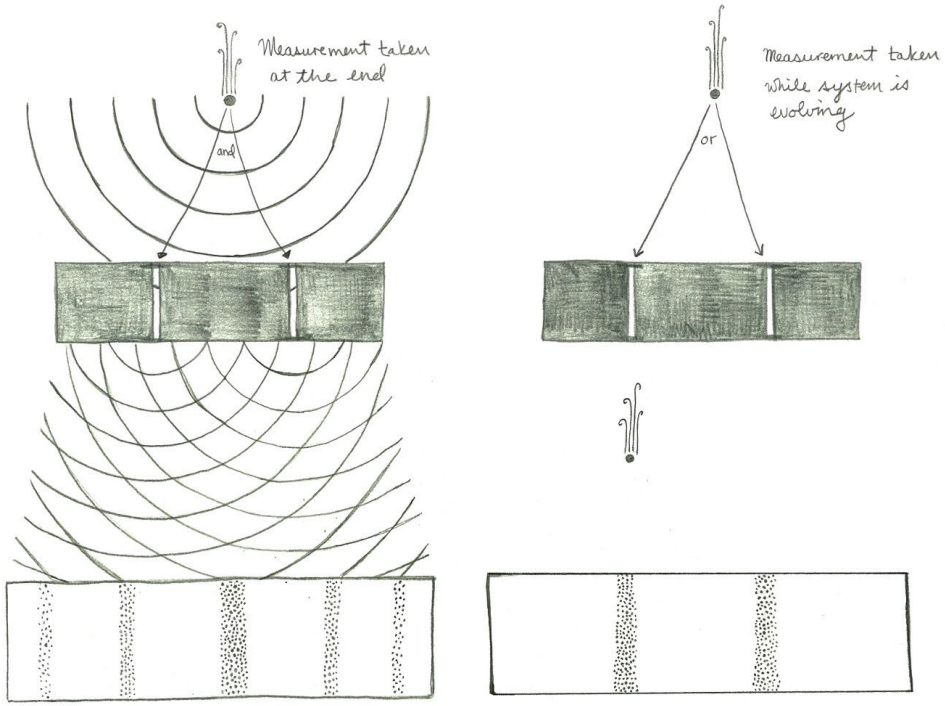
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After the measurement, we know where the electron is, and it cannot be in all the other possible places.

A possible interpretation is that the measurement collapses the wave function!

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The wave function changes due to the measurements.

The equation of Schroedinger



This equation describes the time evolution of the wave function:

$$i\hbar \frac{\partial}{\partial t} \Psi(x,t) = H\Psi(x,t)$$

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Total energy of the system: $\mathrm{H} = \frac{1}{2}mv^2 + V = \frac{p^2}{2m} + V$

The equation of Schroedinger

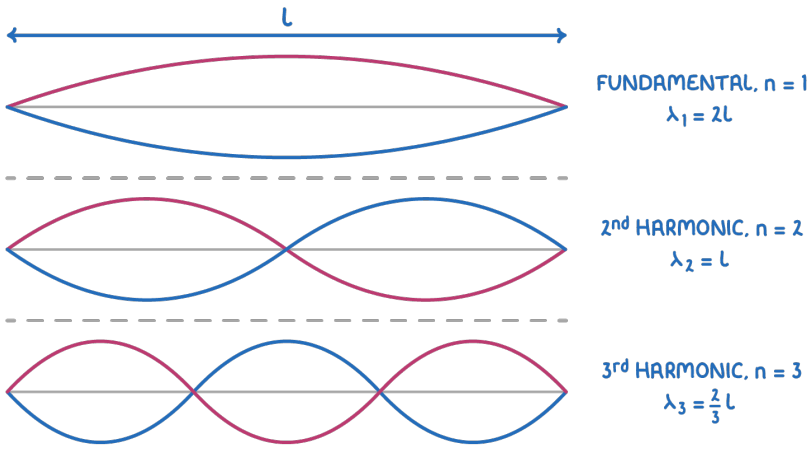


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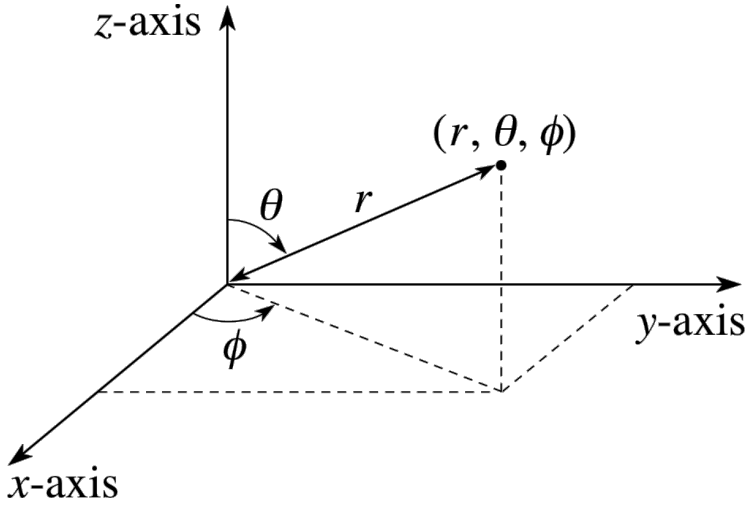


Harmonics are stationary patterns

Time independent solutions of the equation of Shroedinger will obey the following property:

$$H\Psi(x) = E\Psi(x)$$

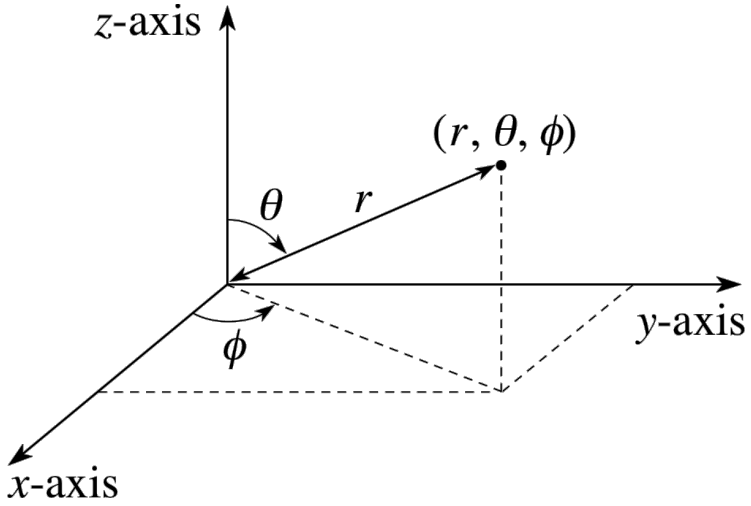
The atom of hydrogen

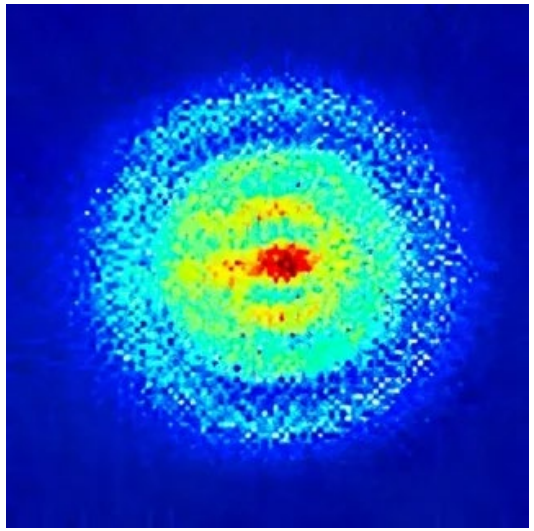


(3D) Equation of Shroedinger for the electron in the hydrogen atom:

$$\left[\frac{-\hbar^2}{2m}\nabla^2 + V(x,y,z)\right]\Psi(x,y,z) = E\Psi(x,y,z)$$

The atom of hydrogen





Stodolna, et al. PRL 2013

(3D) Equation of Shroedinger for the electron in the hydrogen atom:

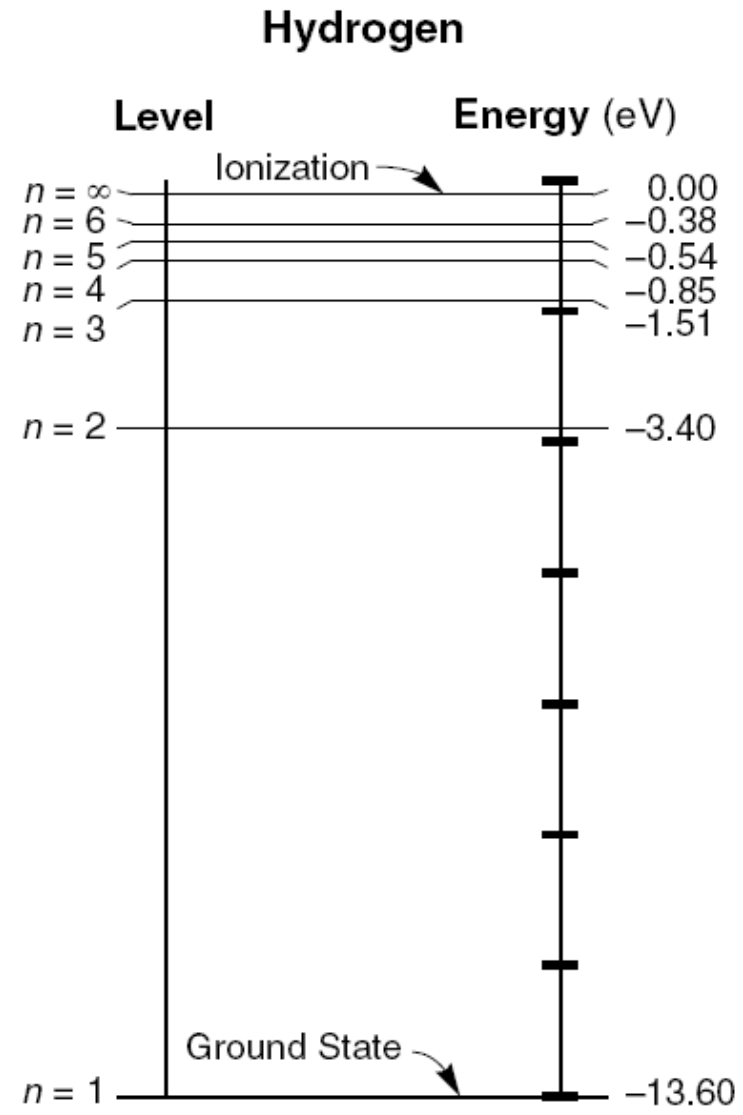
$$\left[\frac{-\hbar^2}{2m}\nabla^2 + V(x,y,z)\right]\Psi(x,y,z) = E\Psi(x,y,z)$$

Since the potential bounding the electron to the nucleus depends only on the distance (Coulomb potential), we can separate the radial part from the angular part of the wave function:

$$\Psi(x,y,z) = R(r)Y(\theta,\phi)$$

$$\left[\frac{-\hbar^2}{2m}\nabla^2 - k\frac{e^2}{r}\right]R(r)Y(\theta,\phi) = ER(r)Y(\theta,\phi)$$

Orbitals



Energy Levels for the Hydrogen Atom

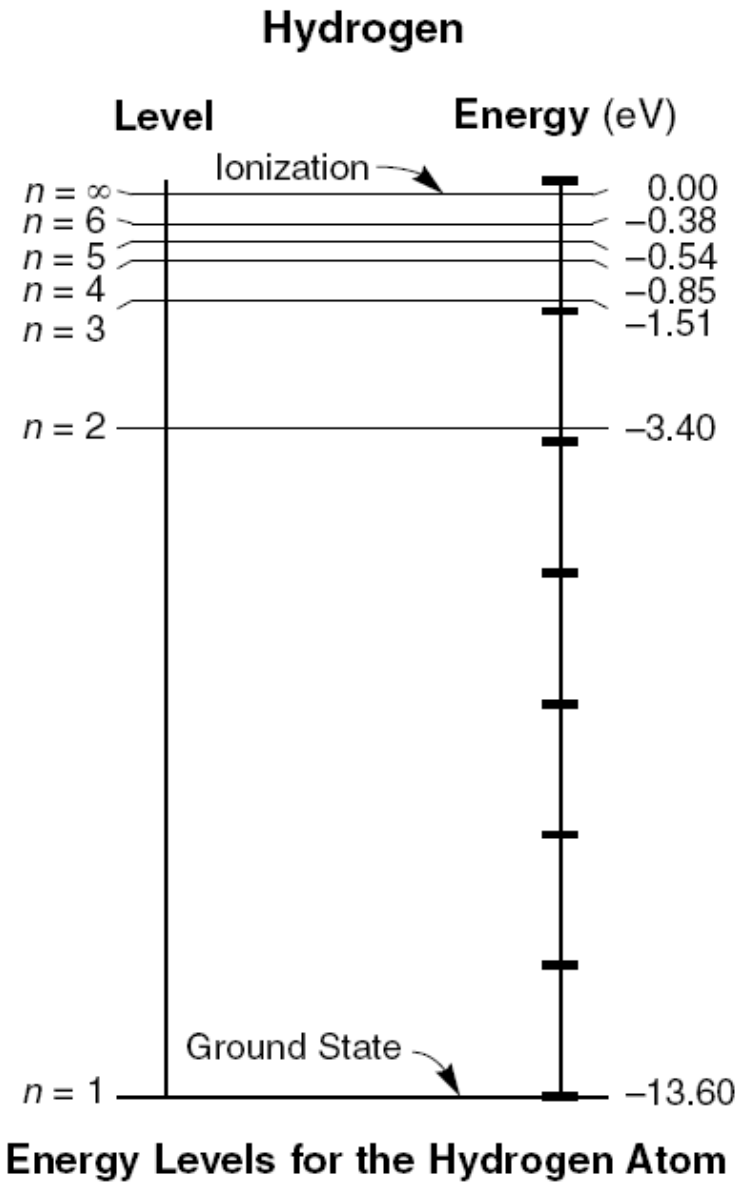
$$HR(r)Y(\theta,\phi) = ER(r)Y(\theta,\phi)$$

The equation of Shroedinger has solution only for discrete values of E.

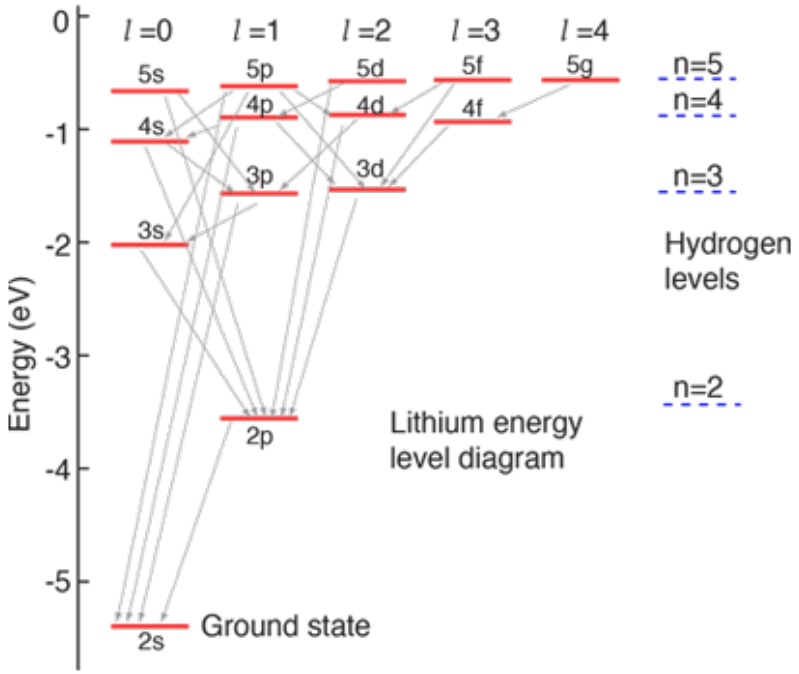
The energy of an electron bound to an atom is quantized.

The solutions give us the energy values, orbital “sizes” and orbital “shapes”.

Orbitals



Energy Levels for the Hydrogen Atom



$$HR(r)Y(\theta,\phi) = ER(r)Y(\theta,\phi)$$

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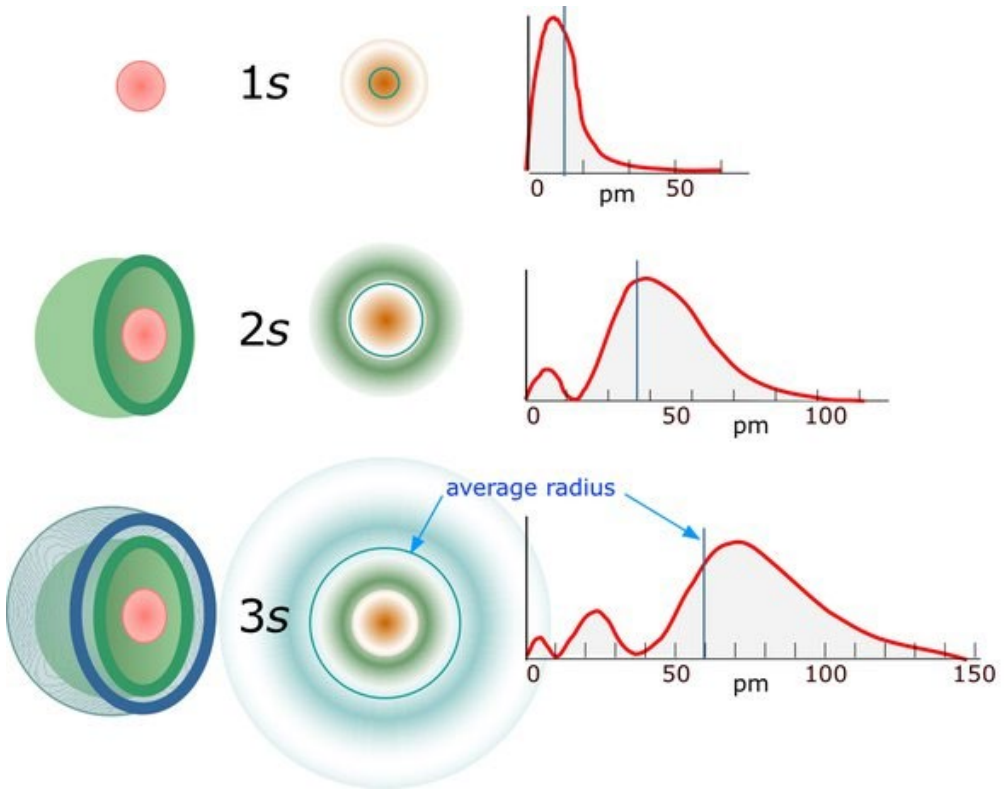
The solutions give us the energy values, orbital "sizes" and orbital "shapes".

For the hydrogen atom, the energy levels do not depend on the angular part of the solution.

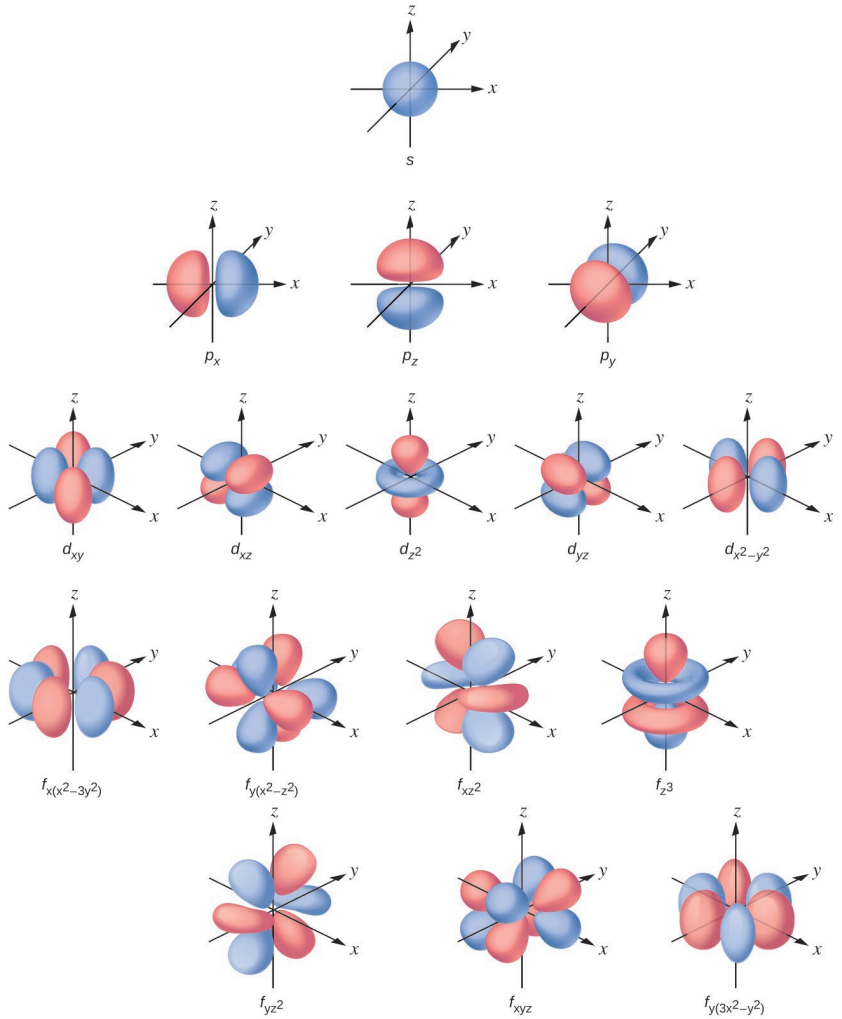
But in general different orbital shapes have different energy.

Orbitals

Orbitals are characterized by few quantum numbers that corresponds to the possible solutions of the Schroedinger equation for that atom.

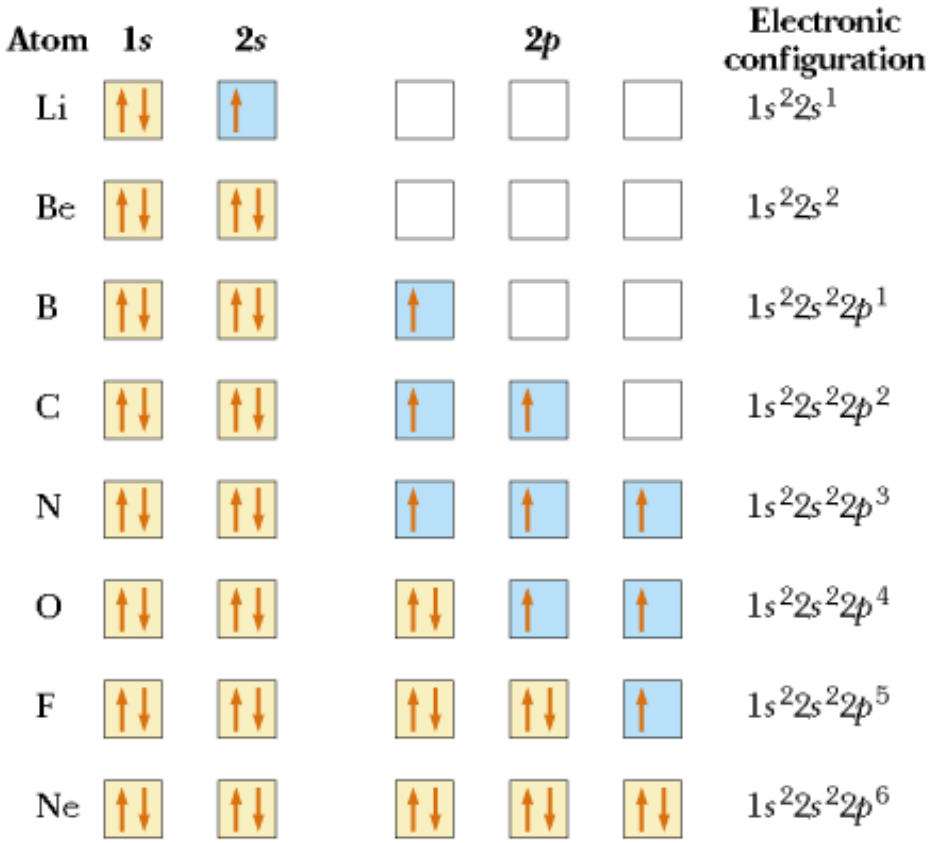


Radial part of the orbital



Angular part of the orbital

Spins and Pauli's exclusion principle



Serway, Physics for Scientists and Engineers, 5/e
Figure 42.14 Harcourt, Inc.

In addition to the quantum numbers that describe an orbital's energy and shape, electrons have an intrinsic quantum property called spin.

This property is quantized and can have one of two values: spin-up (+ 1/2) or spin-down (− 1/2).

Pauli's exclusion principle states that no two electrons in the same atom can have the same set of four quantum numbers.

An atomic or molecular orbital can hold a maximum of two electrons, and these two electrons must have opposite spins.

This explains why electron shells fill up in a specific order, which in turn determines the chemical properties of atoms and the nature of chemical bonds that form molecules.

Molecular orbitals

Molecular orbitals are the solution to the Schroedinger equation for molecules.

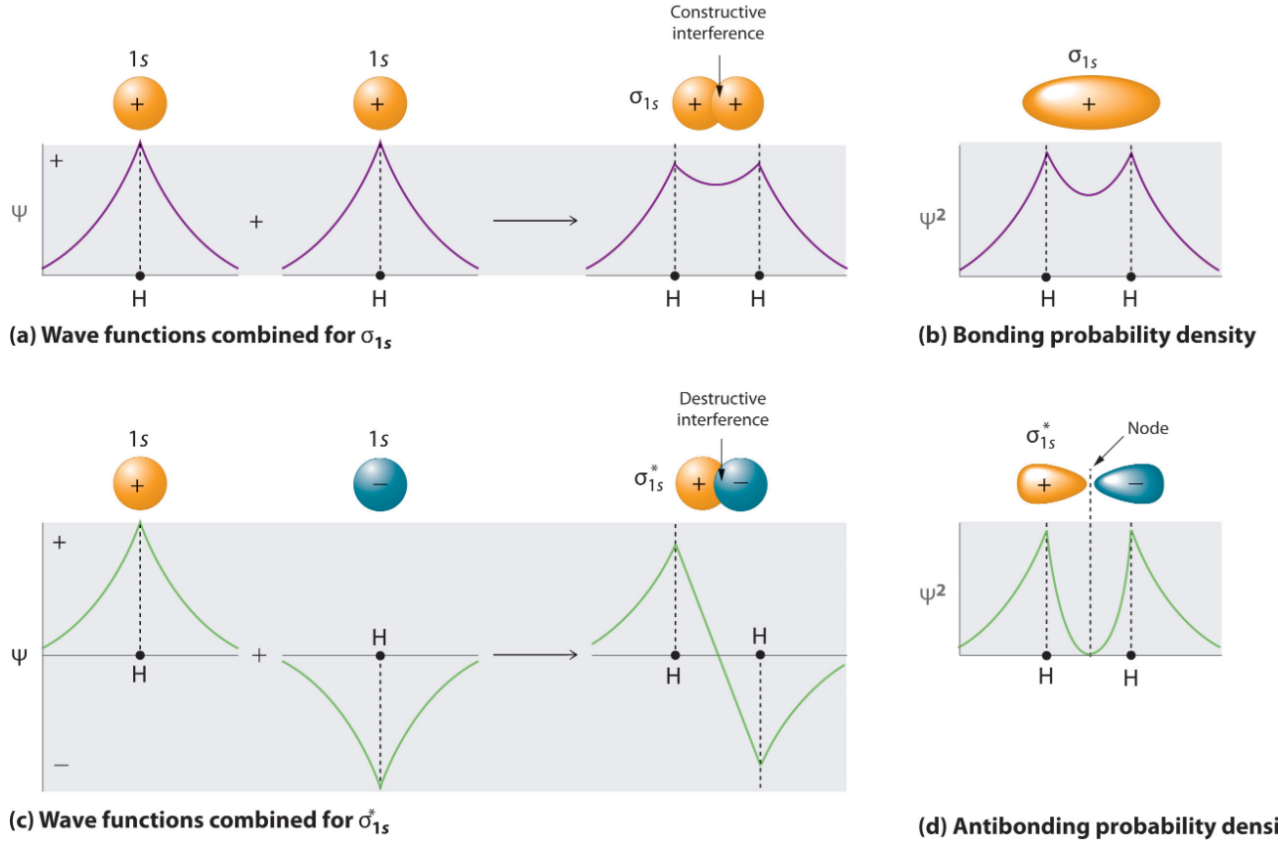


Figure 5.4.1.1: Molecular Orbitals for the H_2 Molecule. (a) This diagram shows the formation of a bonding σ_{1s} molecular orbital for H_2 as the sum of the wave functions (Ψ) of two H 1s atomic orbitals. (b) This plot of the square of the wave function (Ψ^2) for the bonding σ_{1s} molecular orbital illustrates the increased electron probability density between the two hydrogen nuclei. (Recall that the probability density is proportional to the square of the wave function.) (c) This diagram shows the formation of an antibonding σ^*_{1s} molecular orbital for H_2 as the difference of the wave functions (Ψ) of two H 1s atomic orbitals. (d) This plot of the square of the wave function (Ψ^2) for the σ^*_{1s} antibonding molecular orbital illustrates the node corresponding to zero electron probability density between the two hydrogen nuclei.

Molecular orbitals

Energy for excitation: 10 eV

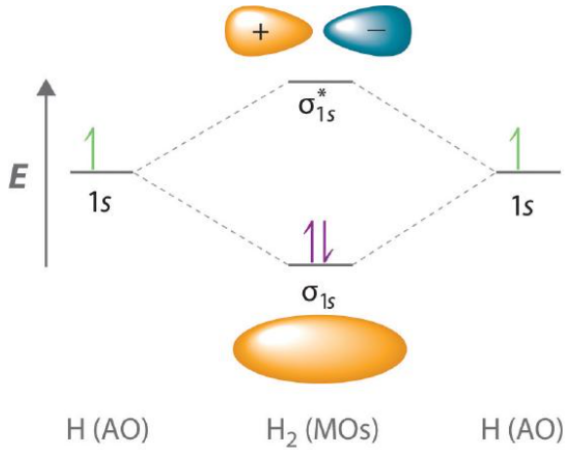
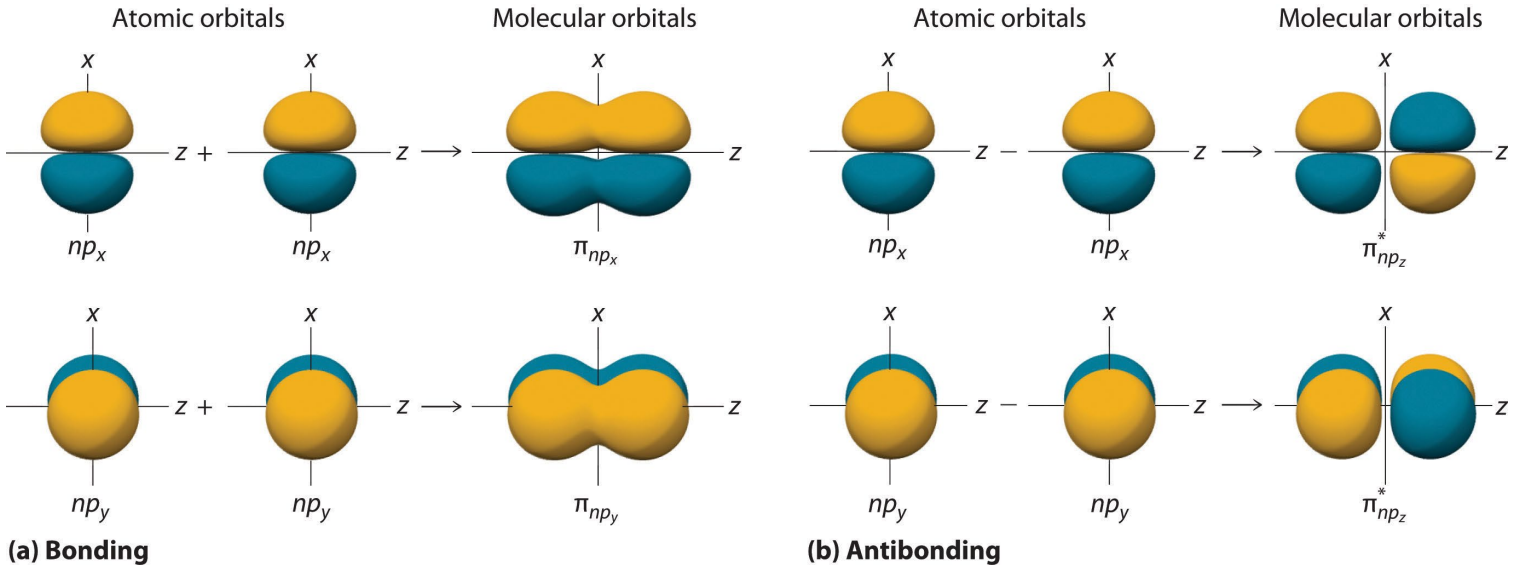
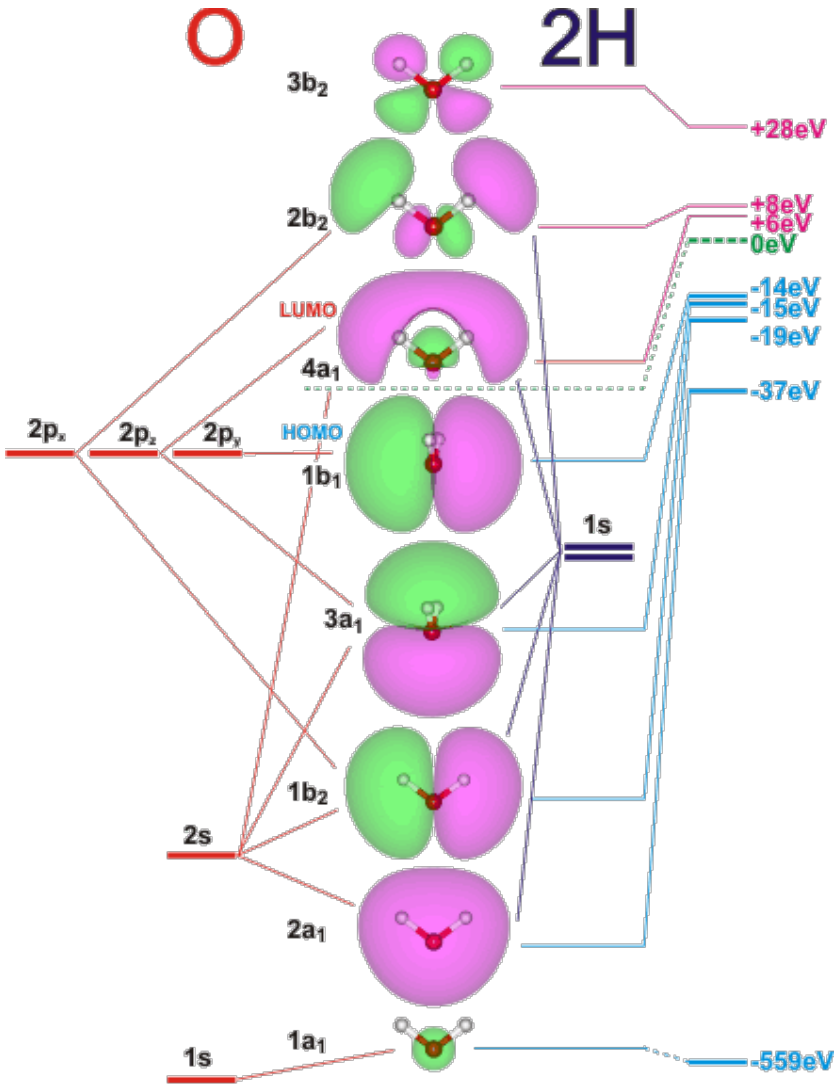


Figure 5.4.1.2: Molecular Orbital Energy-Level Diagram for H_2. The two available electrons (one from each H atom) in this diagram fill the bonding σ_{1s} molecular orbital. Because the energy of the σ_{1s} molecular orbital is lower than that of the two H 1*s* atomic orbitals, the H_2 molecule is more stable (at a lower energy) than the two isolated H atoms.



Electronic orbitals for the water molecule



The molecular orbitals of water are formed from the valence atomic orbitals of the central oxygen atom (2s, 2p) and the two hydrogen atoms (1s).

The specific way these atomic orbitals combine to form molecular orbitals explains water's bent geometry and its unique properties.

The resulting molecular orbitals are not symmetric; the electrons are pulled more toward the highly electronegative oxygen atom.

This uneven distribution of electron density creates the dipole moment that makes water a polar molecule, which is fundamental to its role as the universal solvent in biological systems.